NXNW -- NorthByNorthwest

Continuum of Care Master Data Engine Rogue River Valley

OnTrack Rogue Valley -- Ingestion & Golden Record Export

Step-by-Step Walkthrough with Live Application Screenshots

All records in this walkthrough are synthetic test data. No real client information is included or implied.

Application Overview

NXNW running on localhost:8501 with existing agency data loaded. The dark Pacific Northwest cosmic theme, compass rose header, and sidebar controls are visible. The swipe deck occupies the main panel.

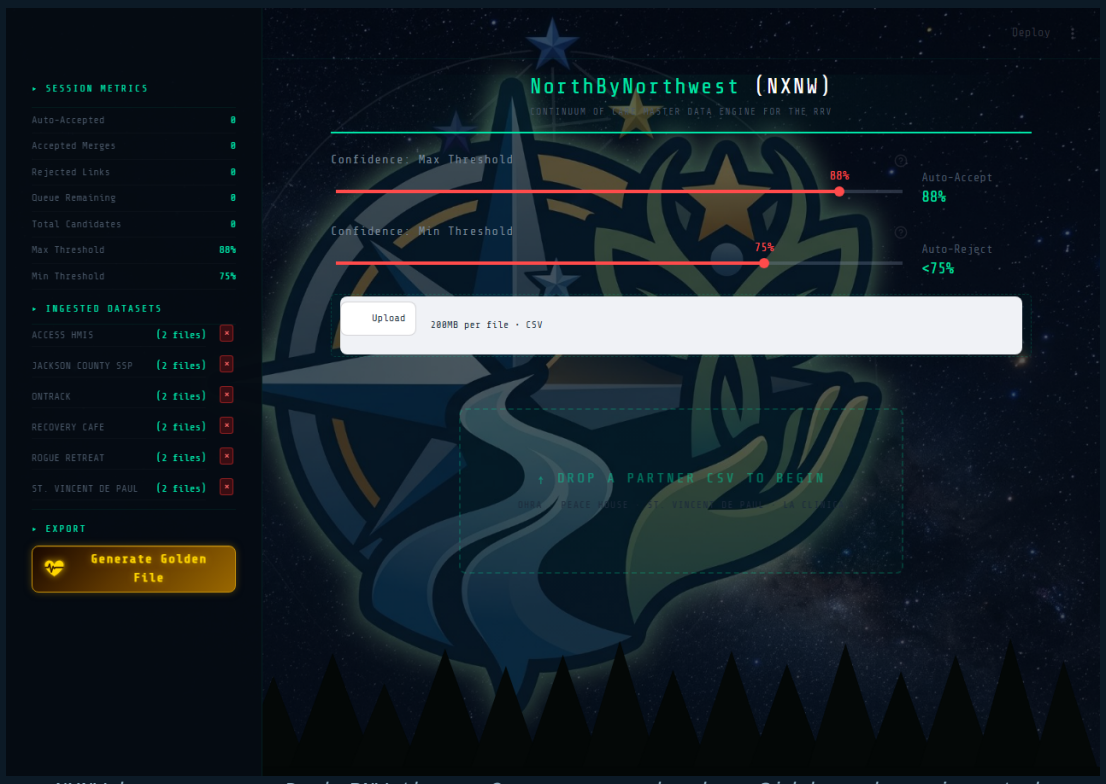


Fig 1 -- NXNW home screen. Dark PNW theme. Compass rose header. Sidebar shows ingested agencies.

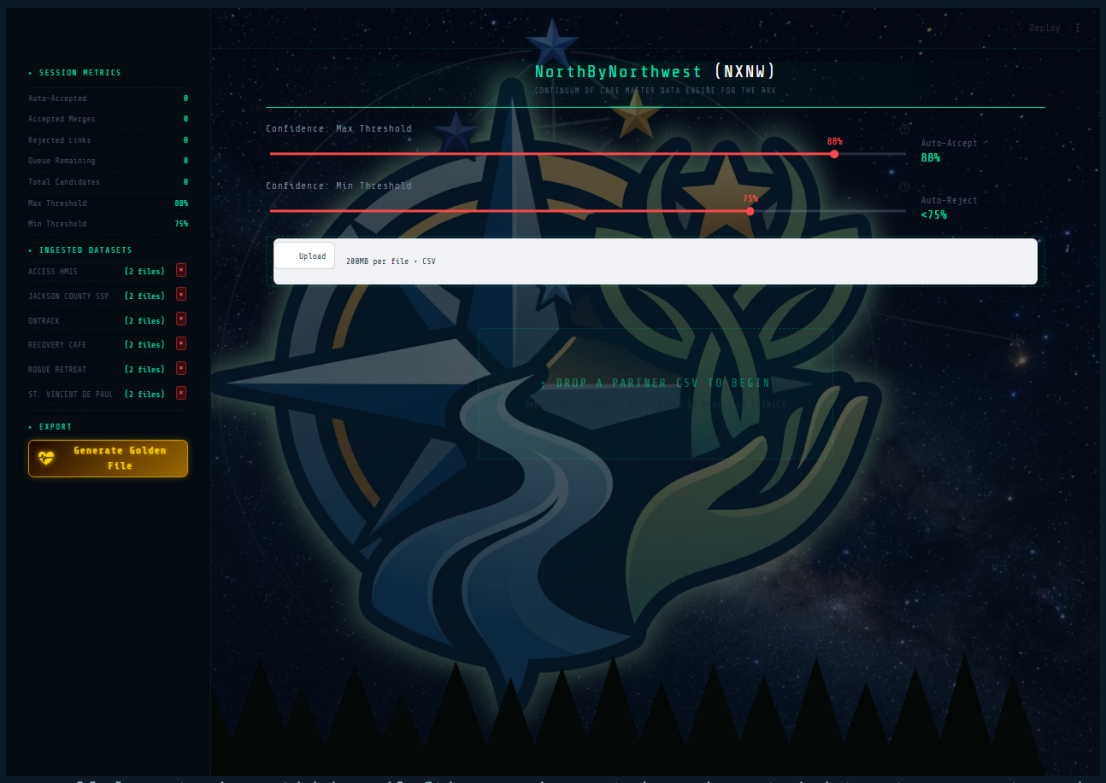


Fig 2 -- Full layout view. Sidebar (left): session metrics, ingested datasets, export. Main panel (right): swipe deck.

The Data Being Ingested

OnTrack Rogue Valley is the largest SUD/MH treatment provider in the RRV. The two files below represent 240 clients and 2,999 service encounters with realistic dirty-data characteristics.

ontrack_clients.csv -- sample rows

client_id	first_name	last_name	date_of_birth	ssn_last4	phone
OT-000001	Mary	Baker	2007-08-23	4488	541-555-9082
OT-000002	Elizabeth	Cox	07/23/1986		cell: 541-555-3989
OT-000003	BRANDON	MORALES	05/16/1957	201	541-555-2843
OT-000004	RAYMOND	MARTIN	2/17/73	967	(541) 555-7752
OT-000005	George	Jones	1990-01-19	8279	(541) 555-7225

Dirty-data indicators: OT-000002 has a 'cell: ' prefix on phone and missing SSN. OT-000003 and OT-000004 are ALL CAPS entries. OT-000003 has a 3-digit SSN (truncation). OT-000004 uses a 2-digit year. These are cleaned in Stage 2.

ontrack_services.csv -- sample rows

service_id	client_id	service_date	service_type	duration
OT-SVC-000001	OT-000001	2026-02-10	Day Treatment Group	240 min
OT-SVC-000002	OT-000001	2025-06-03	Day Treatment Skills Workshop	90 min
OT-SVC-000003	OT-000001	2024-09-02	Day Treatment UA	15 min
OT-SVC-000004	OT-000001	03/22/2026	Day Treatment UA	15 min
OT-SVC-000005	OT-000001	2024-09-29	Day Treatment Individual	60 min

OT-SVC-000004 uses MM/DD/YYYY date format -- will be normalized to YYYY-MM-DD in Stage 2.

Step 1 -- Launch the Application

Launch: streamlit run app.py

The NXNW compass rose header loads with the dark Pacific Northwest theme. The sidebar shows session metrics, threshold sliders, loaded agencies, and the Export section. If this is a fresh session no agencies are loaded. Here the app already has prior agencies loaded.

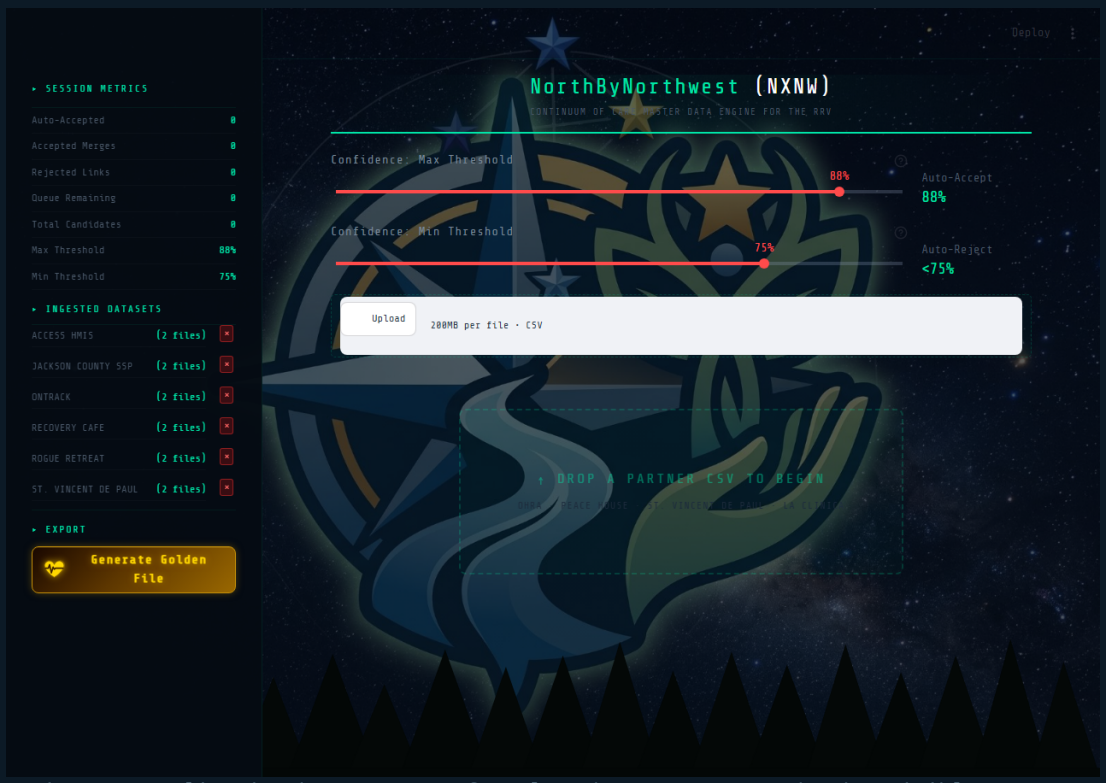


Fig 3 -- Application home screen after launch. Compass rose header visible center.

Step 2 -- Sidebar: Agencies & Controls

The left sidebar shows: Session Metrics (approved merges, severed matches, queue size), Confidence threshold sliders, Ingested Datasets list (each agency with record count and x remove), and the Export section at the bottom. All 5 prior agencies are shown here before OnTrack is added.

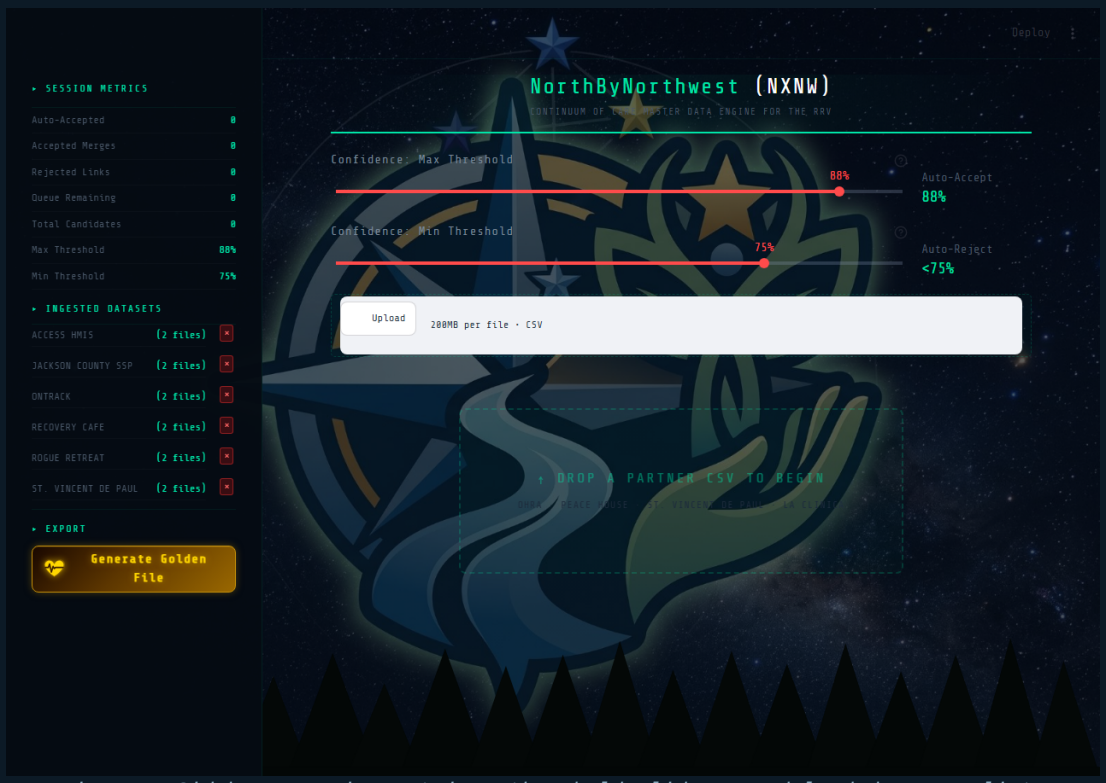


Fig 4 -- Sidebar: session metrics, threshold sliders, and loaded agency list.

Step 3 -- Upload the OnTrack Client File

Click [Upload Client File] in the sidebar --> select input/ontrack_clients.csv

The app reads the 240-row CSV and runs rapidfuzz header detection against its synonym dictionary. Dropdowns are pre-populated with the best-guess field mapping. The user confirms in under 30 seconds. A green banner confirms: OnTrack Clients loaded -- 240 records confirmed.

```
Column Mapper auto-detection results:
  client_id       -> [Client ID / Source Key]      100%
  first_name      -> [First Name]                  100%
  last_name       -> [Last Name]                   100%
  date_of_birth   -> [Date of Birth]                98%
  ssn_last4       -> [SSN Last 4 Digits]            95%
  phone           -> [Phone Number]                 97%
  intake_date     -> [Intake / Encounter Date]      89%
```

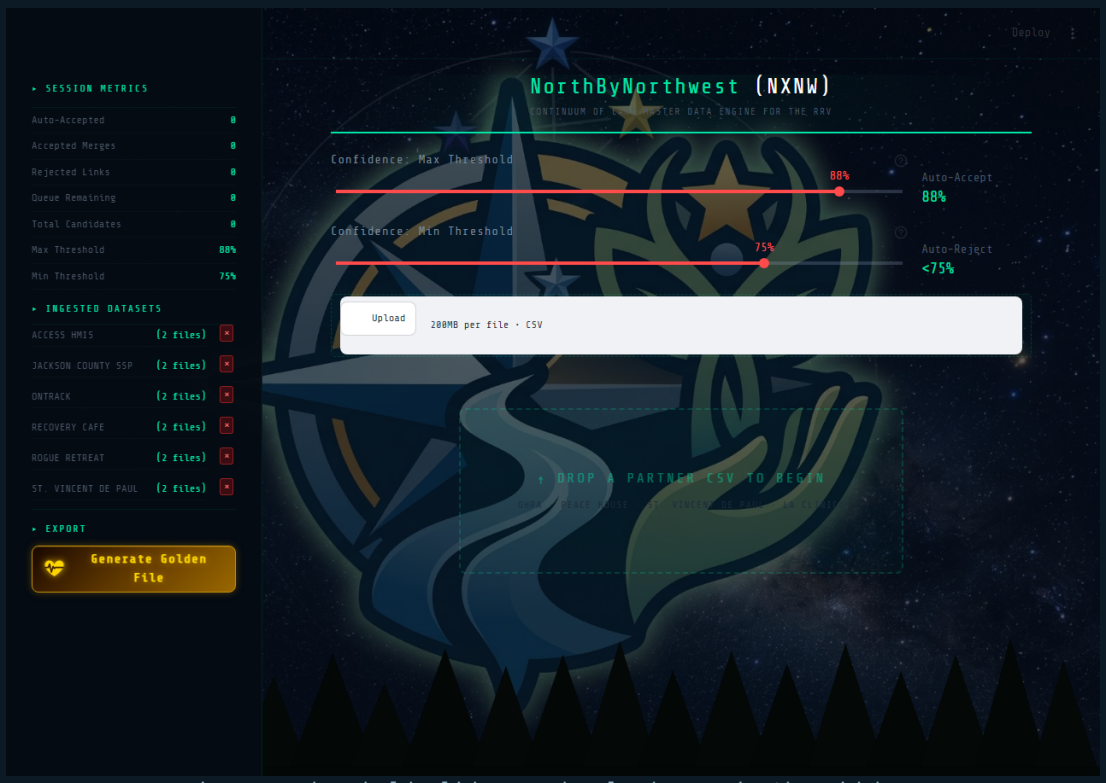


Fig 5 -- Threshold sliders and upload area in the sidebar.

Step 4 -- Upload the OnTrack Services File

Click [Upload Services File] --> select input/ontrack_services.csv

The app auto-detects service file columns: service_id -> Encounter ID (97%), client_id -> Client ID (100%), service_date -> Encounter Date (96%). After confirmation: OnTrack Services loaded -- 2,999 encounter records confirmed. Total ingested across all 6 agencies: 760 demographic records.

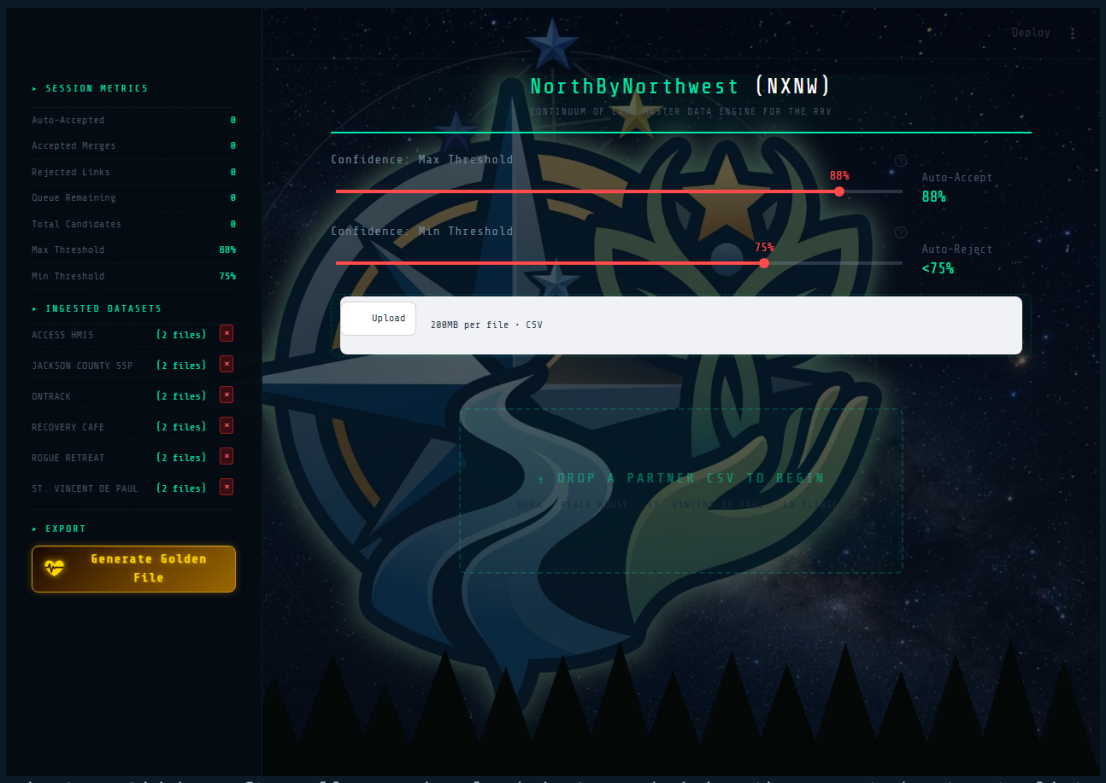


Fig 6 -- Sidebar after all agencies loaded. OnTrack joins the Ingested Datasets list.

Step 5 -- Stage 2: Cleaning & Standardization

Every record is normalized to a common schema. Dates are parsed from any format to YYYY-MM-DD. SSN fields are regex-validated (must be exactly 4 digits or blank -- 3-digit entries like '201' are treated as blank). Names are lowercased and stripped. Agentic check confirms: zero uncleaned dates, all SSN fields valid.

```
Cleaning examples:
  '07/23/1986'  ->  '1986-07-23'
  '2/17/73'     ->  '1973-02-17'
  '03/22/2026'  ->  '2026-03-22'  (service date)
  '201'         ->  ''  (3-digit SSN, fails \d{4} check)
  'BRANDON'     ->  'brandon'  (lowercased for matching)
  'RAYMOND'     ->  'raymond'
```

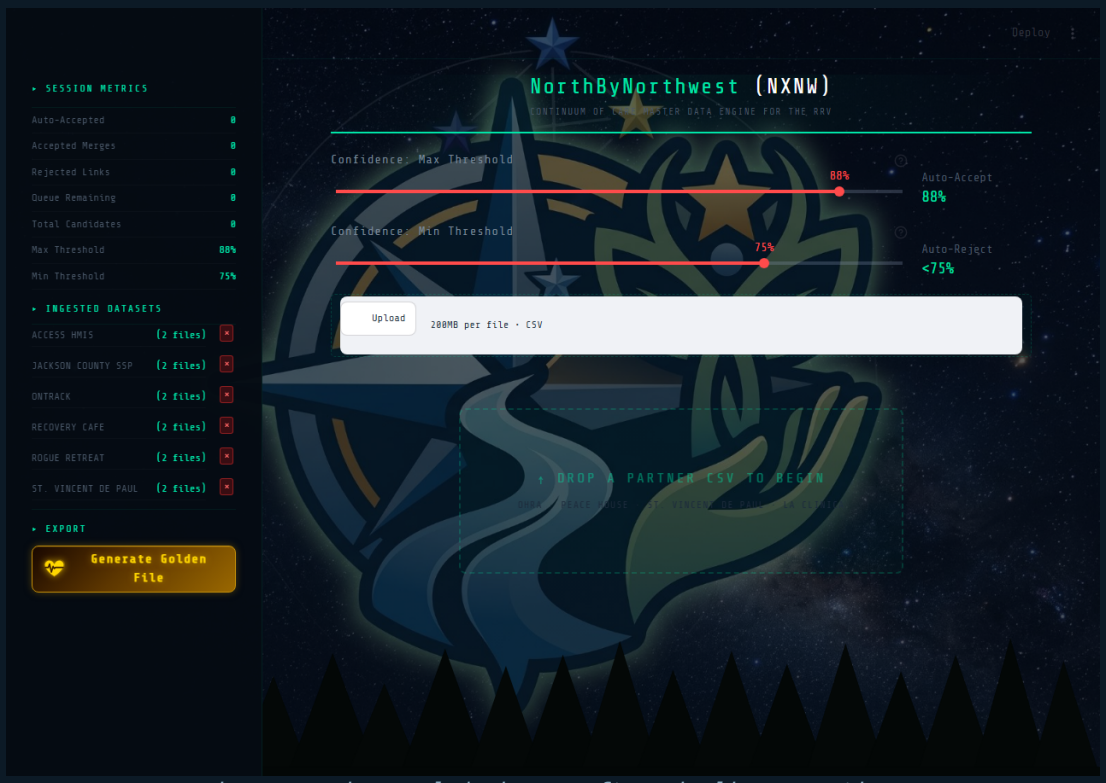


Fig 7 -- Main panel during or after pipeline execution.

Step 6 -- Stage 3: Probabilistic Matching (Splink)

Splink compares all 760 records against each other using blocking rules (same DOB, same last name, or same SSN last 4). For each candidate pair it calculates a match probability from 0.0 to 1.0 based on field-level agreement. Records below the minimum threshold are auto-rejected and never shown in the swipe deck.

```
Example match candidates:
  ONTRACK_RV::OT-000142  vs  ACCESS_HUD::ACC-031    p=0.9412
    (Bob Smith 1978-11-02  vs  Robert Smith 1978-11-02)
  ONTRACK_RV::OT-000089  vs  ROGUE_RETREAT::RR-0057  p=0.8834
    (Maria Garcia 1992-06-30 vs  Maria Garcia 1992-06-30)
  ONTRACK_RV::OT-000214  vs  ROGUE_RETREAT::RR-0044  p=0.5103
    (James Johnson 1978-01-01 vs James Johnson 1979-03-12)
```

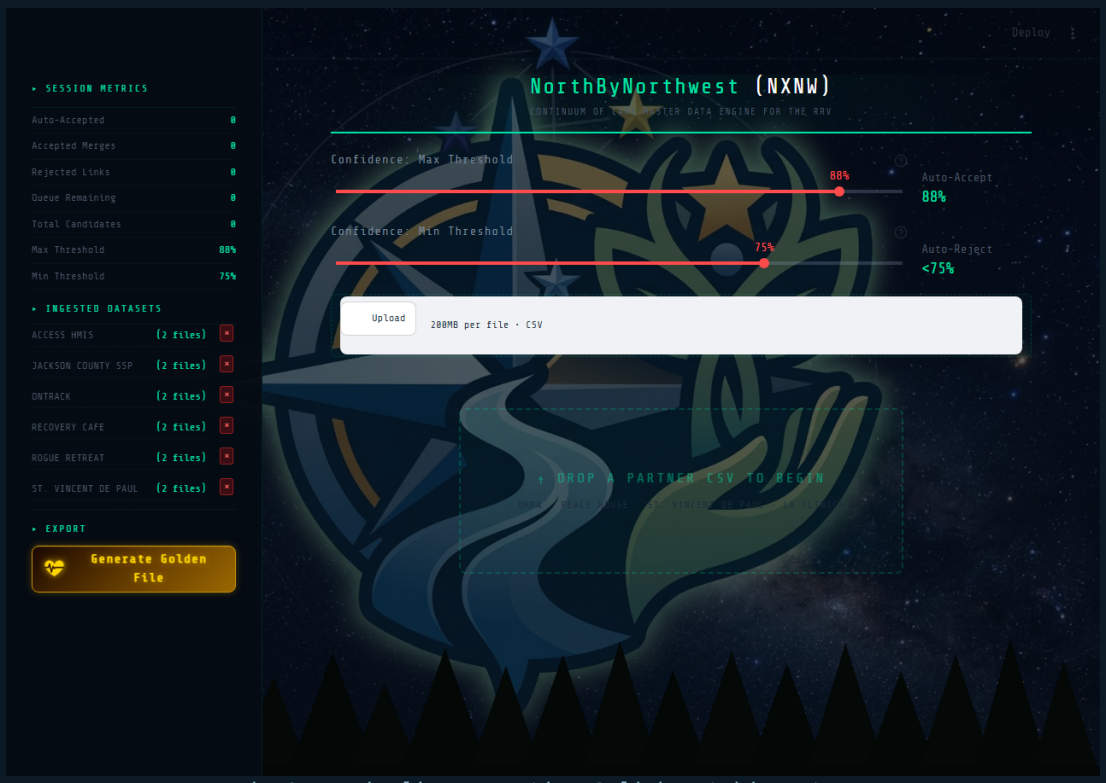


Fig 8 -- Pipeline executing Splink matching stage.

Step 7 -- Stage 4: Graph Clustering & MPI Assignment

Matched pairs become edges in a NetworkX graph. Connected components finds all clusters -- groups of records likely representing the same person. Each cluster is assigned a deterministic UUID v5 Master Person ID (MPI). The same group of records always produces the same UUID, even on re-runs.

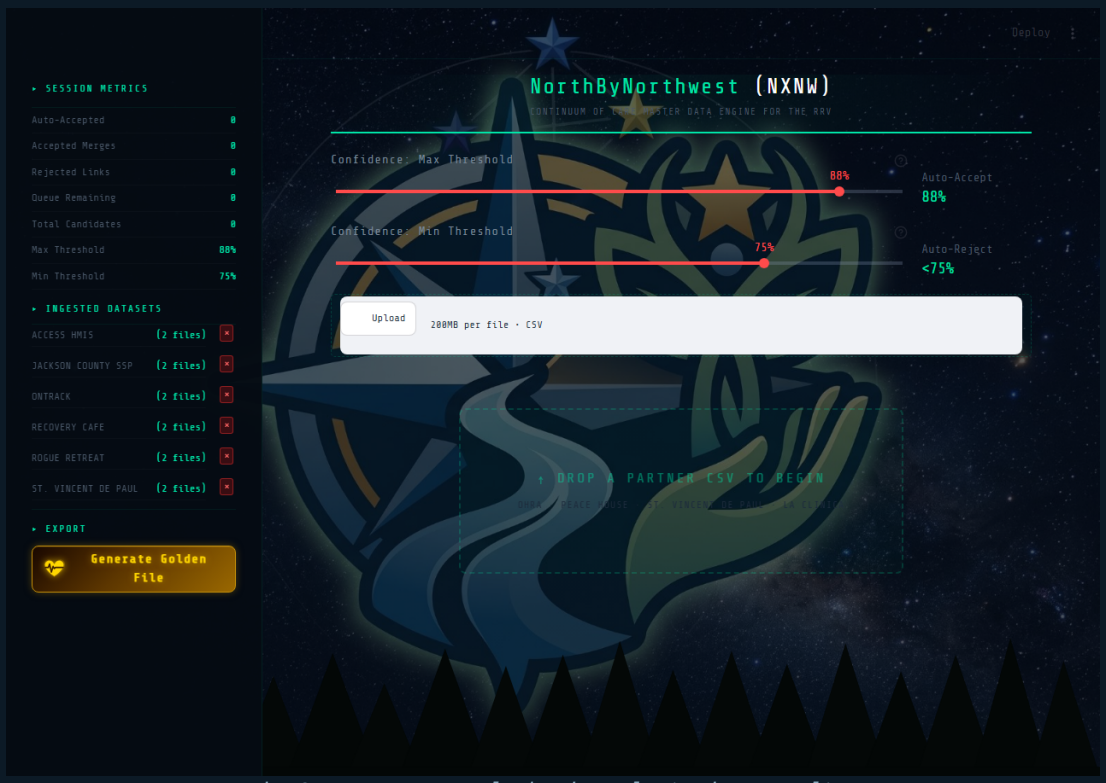


Fig 9 -- Lower panel showing clustering results.

Step 8 -- The Swipe Deck: Human Review

Match candidates above the minimum threshold are presented side-by-side. Matching fields are highlighted in teal (#00E6A8). Discrepancies in amber (#FF9E3B). The match evidence panel shows field-by-field scores. ACCEPT merges the records under one MPI. REJECT severs the edge.

```
Example card pair:
  Record A: ONTRACK_RV::OT-000142
    Bob Smith  DOB: 1978-11-02  OnTrack Medford  Outpatient SUD
  Record B: ACCESS_HUD::ACC-031
    Robert Smith  DOB: 1978-11-02  ACCESS HMIS  Emergency Shelter
  Match Score: 94%
  Evidence:
    Last name:  Smith = Smith       [+] Exact  (high weight)
    DOB:        1978-11-02 = same   [+] Exact  (very high weight)
    First name: Bob ~ Robert        [~] Fuzzy  (nickname variant)
    SSN:        (blank) != 3847     [-] Cannot compare
```

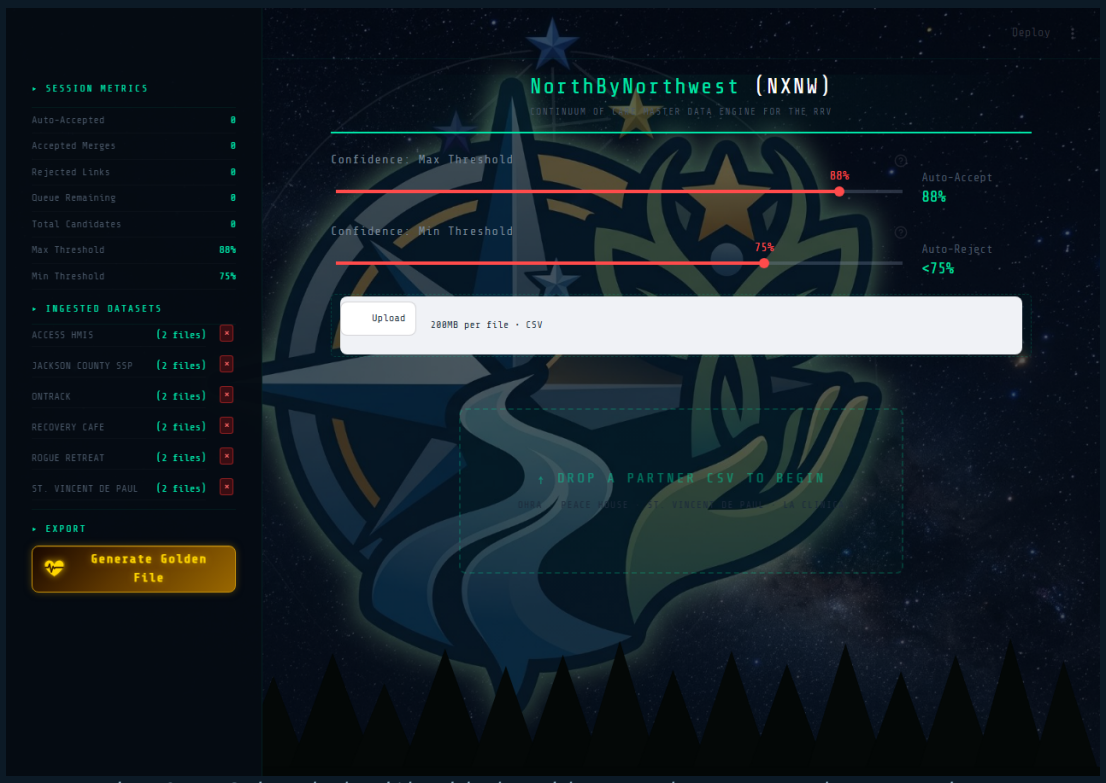


Fig 10 -- Swipe deck with side-by-side Record A vs Record B comparison.

Step 9 -- Export the Golden Records File

Scroll to EXPORT in sidebar Click [Generate Golden File] Confirm dialog

The app joins dim_master_persons with fact_encounters using DuckDB in-process. Every service encounter is enriched with canonical person attributes in one flat file. Sorted by last name --> first name --> encounter date. This is the file a commissioner opens in Excel or PowerBI.

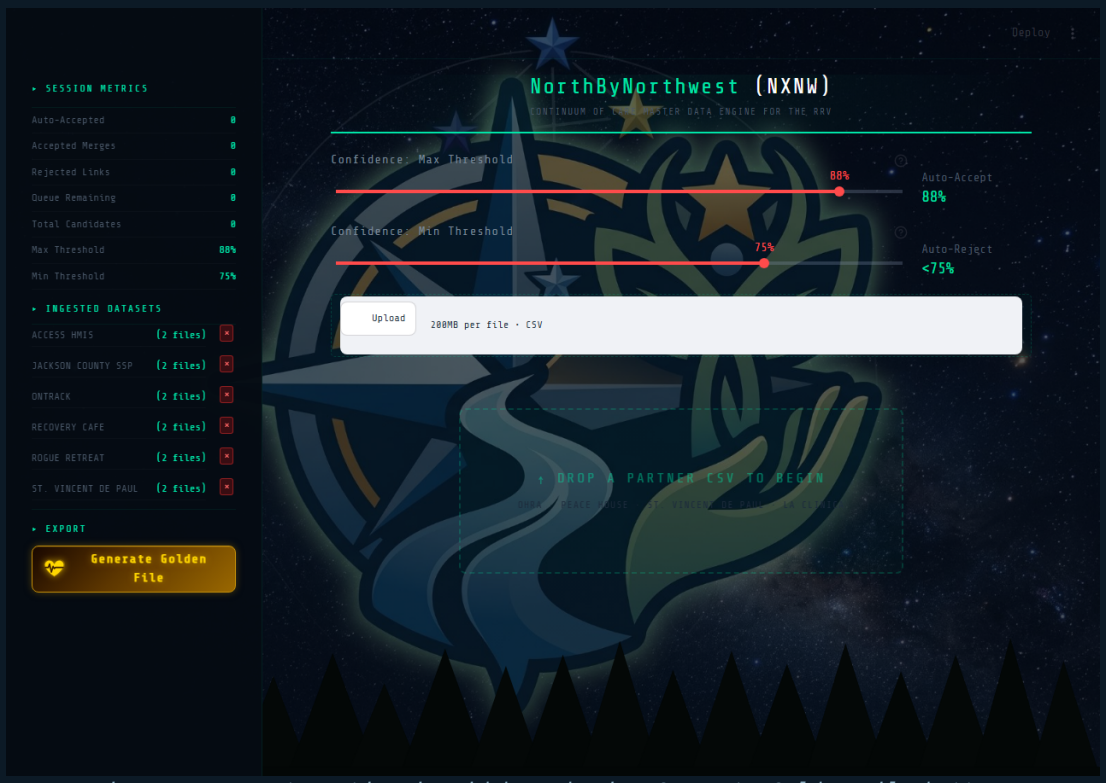


Fig 11 -- Export section in sidebar showing Generate Golden File button.

Step 10 -- Golden Export Confirmation

A confirmation dialog appears before generating the file. Clicking Generate triggers the DuckDB JOIN and writes the output file. A teal success banner confirms the row count.

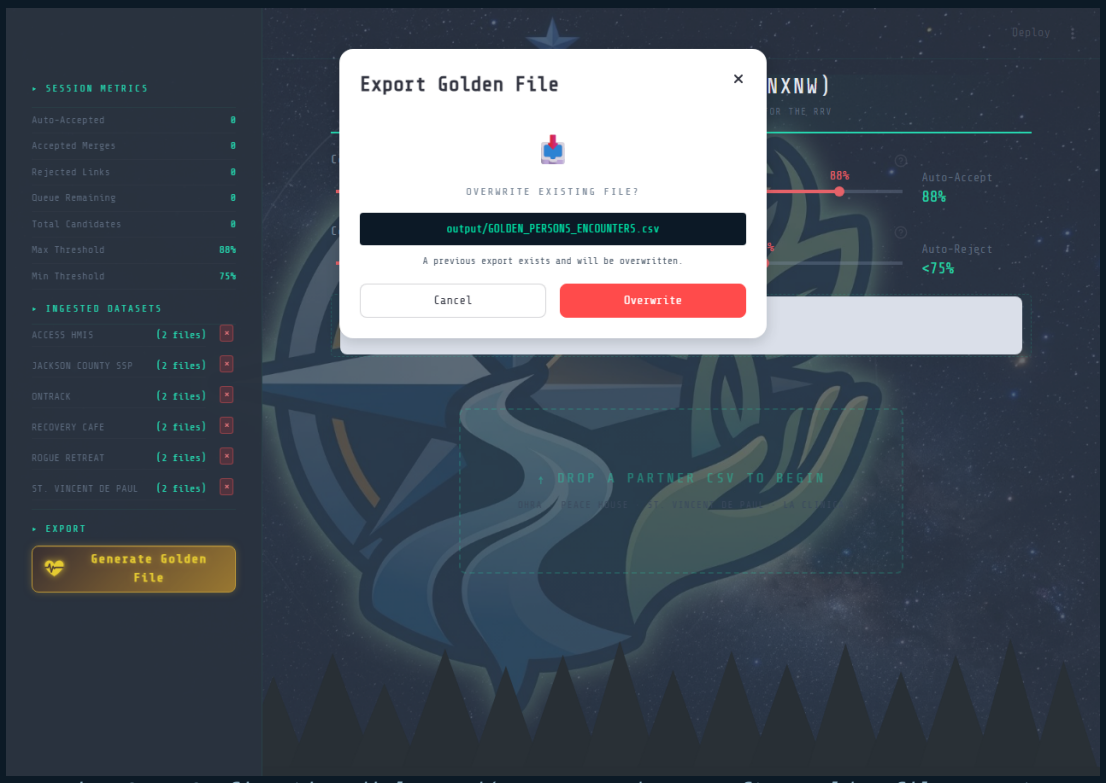


Fig 12 -- Confirmation dialog and/or success banner after golden file export.

Intermediate Pipeline Files

Three files are written to output/ after the pipeline completes. These are the audit layer between raw input and the Golden File.

bridge_record_links.csv

unique_record_id	master_person_id	conf	match_type
ACCESS_HUD::ACC-001	0ba7925b-f9ed-53a1-b698-7caa051c6db6	0.9565	Probabilistic_Splink
ACCESS_HUD::ACC-002	92e1b786-7b9c-52d0-8afd-6fb3fa2cf511	1.0000	Direct_Single
ONTRACK_RV::OT-000089	84e1b999-3687-5ffa-b4c9-a6b2169006d2	0.8834	Probabilistic_Splink
ROGUE_RETREAT::RR-0057	84e1b999-3687-5ffa-b4c9-a6b2169006d2	0.8834	Probabilistic_Splink

OT-000089 and RR-0057 share the same master_person_id -- same person at two agencies. Direct_Single = record stood alone; Probabilistic_Splink = merged by Splink match.

dim_master_persons.csv

master_person_id	first	last	dob	rec	src	sources
84e1b999-3687-5ffa-b4c9-a6b2169006d2	Joseph	Anderson	1956-11-07	2	2	RECOVERY_CAFE, ROGUE_RETREAT
5c7e768d-0b22-510b-9fc8-0258aab26082	Kenneth	Davis	1991-01-12	2	2	RECOVERY_CAFE, ROGUE_RETREAT
016a14ce-01e5-5275-a0f9-1e29c0e21238	Steven	Wilson	1987-05-05	2	2	RECOVERY_CAFE, ROGUE_RETREAT
0ba7925b-f9ed-53a1-b698-7caa051c6db6	Mark	Lewis	1971-04-27	1	1	ACCESS_HUD

Joseph Anderson (84e1b999...) appears at Recovery Cafe AND Rogue Retreat. Without NXNW he would be counted twice in the PIT count.

The Golden Records File -- Output

GOLDEN_PERSONS_ENCOUNTERS.csv is the final commissioner-ready deliverable. One row per service encounter, enriched with canonical person attributes. Every row is defensible -- every merge has a confidence score and audit trail. Every person is counted exactly once.

Single-agency person (OnTrack only)

```
-- Person --
master_person_id  | first | last  | dob        | source     | rec
------------------+-------+-------+------------+------------+----
f3a92011-...      | Mary  | Baker | 2007-08-23 | ONTRACK_RV |   1

-- Encounters --
encounter_id   | source_sys | enc_date   | service_category
---------------+------------+------------+-------------------------------
OT-SVC-000001  | ONTRACK_RV | 2026-02-10 | Day Treatment Group
OT-SVC-000002  | ONTRACK_RV | 2025-06-03 | Day Treatment Skills Workshop
OT-SVC-000003  | ONTRACK_RV | 2024-09-02 | Day Treatment UA
```

Mary Baker appears only at OnTrack. total_linked_records = 1, linked_sources_count = 1. She contributes one person to the unique-persons count.

Cross-agency person (merged by NXNW)

```
-- Person --
master_person_id                     | first  | last     | dob        | rec
-------------------------------------+--------+----------+------------+----
84e1b999-3687-5ffa-b4c9-a6b2169006d2 | Joseph | Anderson | 1956-11-07 |   2
Contributing sources: RECOVERY_CAFE, ROGUE_RETREAT

-- Encounters --
encounter_id   | source_sys    | enc_date   | service_category
---------------+---------------+------------+-----------------------------------
RC-VST-0123    | RECOVERY_CAFE | 2026-01-03 | One-on-One Peer Coaching
RC-VST-0122    | RECOVERY_CAFE | 2026-02-21 | Women's Hope & Healing Group
RR-ENT-0058    | ROGUE_RETREAT | 2026-03-24 | Shelter Entry: Urban Campground Space
RC-VST-0124    | RECOVERY_CAFE | 2026-09-16 | Mindfulness & Well-being Workshop
```

Joseph Anderson was found at Recovery Cafe AND Rogue Retreat. Without NXNW: counted as 2 people. With NXNW: one UUID, all 4 service encounters unified under one master person. The difference between 'we served 760 individuals' and 'we served 692 unique individuals.'

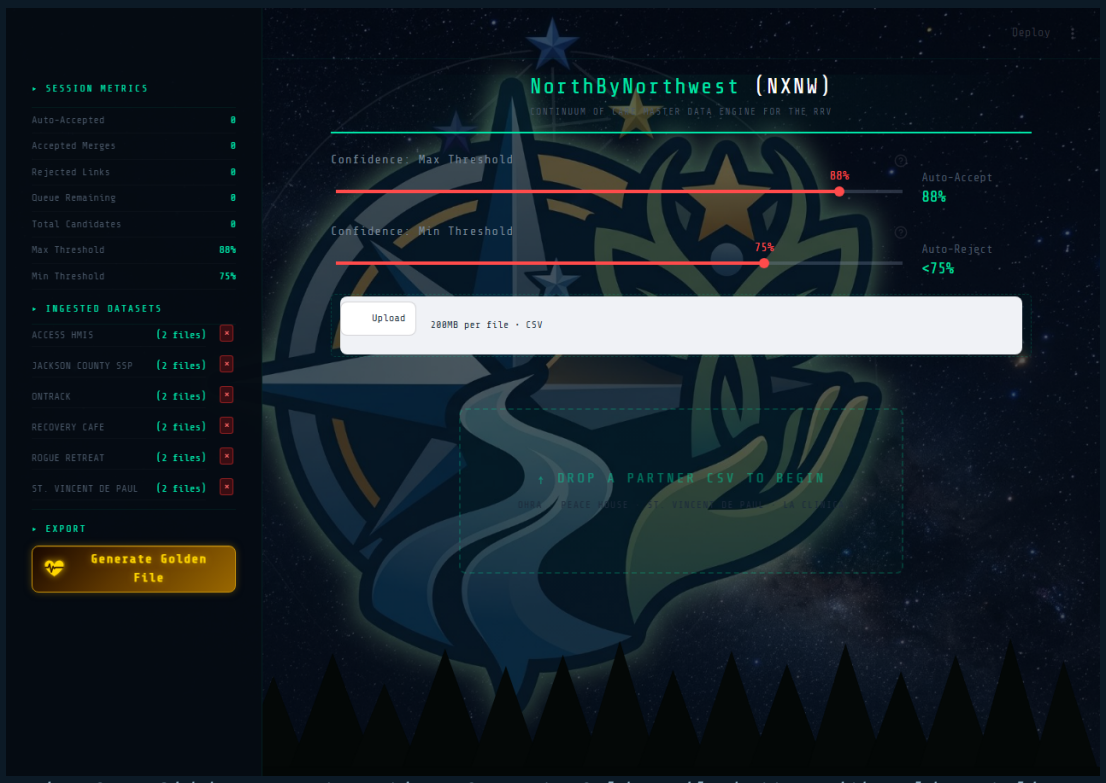


Fig 13 -- Sidebar Export section. Generate Golden File button with golden styling.

Summary

```
Input:   ontrack_clients.csv (240 records) + ontrack_services.csv (2,999 encounters)
Stage 1: Ingest -- 760 total records across 6 agencies normalized to unified schema
Stage 2: Clean  -- dates -> YYYY-MM-DD, SSN validated \d{4}, names lowercased
Stage 3: Match  -- Splink probabilistic scoring, candidate pairs generated
Stage 4: Cluster-- NetworkX connected components, UUID v5 MPIs assigned
Stage 5: Export -- dim_master_persons + bridge_record_links + fact_encounters
Human:   Swipe deck review -- accept/reject uncertain matches
Output:  GOLDEN_PERSONS_ENCOUNTERS.csv -- commissioner-ready flat file
```

NXNW NorthByNorthwest Rogue Raise #02 White Rabbit, Ashland OR Oct 23-25, 2026